

MAX-CUT: choose $S \subseteq V$ st. $|E(S, \bar{S})|$ is maximum (NP-hard, reduction from 3-SAT)

- | | |
|--------------------------|---|
| 1 Local search | } 2-approximation
(best approximation factor is $\frac{1}{0.878} \sim 1.14$)
by Goemans-Williamson |
| 2 Greedy | |
| ✓ 3 <u>Randomisation</u> | |

Using 1-2-3 as approaches is difficult to get better approx factor 2
as our UB for OPT is $|E|$

For example, consider $G = K_{2n}$ = complete graph with $|V| = 2n$ nodes



$$|S| = i \Rightarrow |E(S, \bar{S})| = i \times (2n - i) \stackrel{\substack{\text{max when } i = n \text{ (two factors} \\ \text{are equal)}}}{=} n^2 = \text{OPT}$$

$|S|$ neighbours in $\bar{S}$ of any node in S

approximation based on

$$\text{UB} = |E| \Rightarrow \frac{|E|}{\text{cost}(S)} \leq 2 : \text{ here } |E| = \frac{2n(2n-1)}{2} \sim 2n^2$$

$$\Rightarrow \text{cost}(S) \geq \frac{|E|}{2} \sim n^2$$

3 Randomization

As we saw in the last class, MAX-CUT can be as 2-coloring of the nodes

Randomized 2-coloring

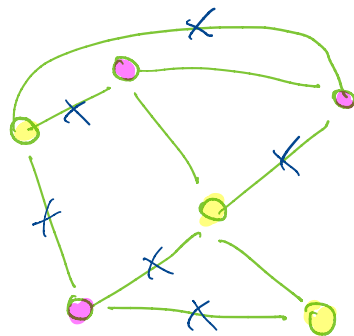
3.a $S = \{y\}$

3.b for each node $u \in V$ do: assign u to S with $\text{pr} = \frac{1}{2}$

Implicitly, for any edge ij we have that

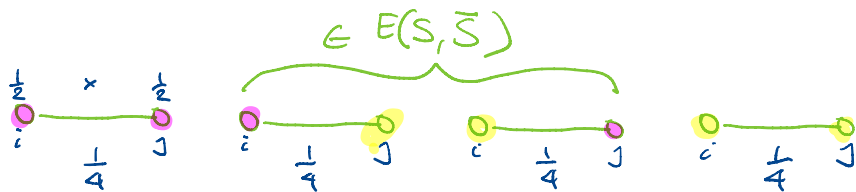
$ij \in E(S, \bar{S})$ iff $\text{color}(i) \neq \text{color}(j)$

$$\text{node } u: \text{color}(u) = \begin{cases} \text{pink} & u \in S \\ \text{yellow} & \text{o.w. } (u \in \bar{S}) \end{cases}$$



$$\text{pr}(u \in S) = \frac{1}{2} \text{ by construction (step 3.b)}$$

$$\Pr(ij \in E(S, \bar{S})) = \frac{1}{2}$$



[obs. using any p instead of $\frac{1}{2}$: $p^2, p(1-p), (1-p)p, (1-p)^2$)]

$\Pr(ij \in E(S, \bar{S})) = 2p(1-p) \leftarrow \text{max when } p = (1-p) \Rightarrow p = \frac{1}{2}$

$$X_{ij} = \begin{cases} 1 & ij \in E(S, \bar{S}) \\ 0 & \text{o.w.} \end{cases} \quad \Pr = \frac{1}{2} \quad \Rightarrow E[X_{ij}] = \frac{1}{2}$$

$$|E(S, \bar{S})| = \sum_{\substack{ij \in E \\ i < j}} X_{ij} \Rightarrow E[|E(S, \bar{S})|] = \sum_{\substack{ij \in E \\ i < j}} \underbrace{E[X_{ij}]}_{\text{"1/2"}} = \frac{|E|}{2} \quad \square$$

random variable
as it depends on
the random 2-coloring

[obs Using any other $p \neq \frac{1}{2}$ does not improve over this by (*)]

Derandomization: or remove dependency on random choice

MAX-CUT is a running example for $\rightarrow$ universal hash functions $\rightarrow$ DETERMINISTIC ALGORITHM
 $\searrow$ conditional expectations

TRIVIAL DERANDOMIZATION: instead of making random choices in a given domain D , just go through all possible choices in D , and examine each of them.

- i) $\text{best_cutsize} = 0$
 - ii) for each possible 2-coloring of the n nodes
 - compute the cutsize S for the given 2-coloring
 - if $S > \text{best_cutsize}$ then best_cutsize = S
 - iii) return best_cutsize
- $\left. \begin{array}{l} \text{ii)} \\ \text{iii)} \end{array} \right\} 2^n \text{ times as there are so many 2-colorings of the input graph } G$

TOTAL TIME: $O(2^n \text{poly}(n))$

First technique: use universal hash family $\mathcal{H}$, where $h \in \mathcal{H}$
 $h: V \rightarrow [m]$, where $m=2$ as we want a 2-coloring

$$(*) \text{ color}(u) = \begin{cases} \text{pink} & \text{if } h(u) = 1 \\ \text{yellow} & \text{o.w (if } h(u) = 0) \end{cases}$$

How can we apply this? Try to obtain the same pr $(ij \in E(S, \bar{S})) = \frac{1}{2}$
but using a space of $O(n^2)$ 2-colorings instead of 2^n ones

IDEA Let k such that $2^{k-1} < n \leq 2^k$, where $n = |V|$ and $V = [n]$ wlog
 $h \in \mathcal{H} : h(x) = ((2x + b) \% p) \% m$ where $p > 2^k$, $m = 2$

Instead of 2-coloring each node, we make a random choice of $h \in \mathcal{H}$
 and define the color of node u as in $(*)$

Because of this setup, we get

$$\Pr[ij \in E(S, \bar{S})] = \Pr[h(i) \neq h(j)] = 1 - \underbrace{\Pr[h(i) = h(j)]}_{\substack{\text{"} \\ \frac{1}{m} \text{ by def. } H \\ \text{i.e. } \frac{1}{2}}} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Pr[X_{ij} = 1]$$

Recall $h \in H$ is equivalent to choosing $a, b \in \mathbb{Z}_p$ where $p = \Theta(2^k)$ and $a \neq 0$
 $\Rightarrow$ all possible choices of $h \in H$ are all possible choices of $a, b \in \mathbb{Z}_p$
 $\Rightarrow O(2^k \cdot 2^k) = O(n^2)$ by our choice of k

- i) $\text{best_cutsize} = 0$
- for each possible $h \in H$:
- ii) for each possible 2-coloring of the n nodes
 - compute the cutsize s for the given 2-coloring
 - if $s > \text{best_cutsize}$ then $\text{best_cutsize} = s$
- iii) return best_cutsize

$\Updownarrow$
it is DETERMINISTIC

In this way, we reduce the number of 2-colorings from 2^n to $O(n^2)$, still preserving $\Pr = \frac{1}{2}$ for an edge being part of the cutset

Second technique: Conditional expectations

IDEA Suppose we can evaluate conditional expectation $E[X|Y]$

$$\begin{aligned} E[X] &= p \cdot \underbrace{E[X|Y]}_a + (1-p) \underbrace{E[X|\bar{Y}]}_b & p &= \text{pr}(Y) \\ & & 1-p &= \text{pr}(\bar{Y}) \\ &= p \cdot a + (1-p) \cdot b \leq p \cdot \max\{a, b\} + (1-p) \cdot \max\{a, b\} = \max\{a, b\} \end{aligned}$$

Hence, $E[X] \leq \max\{E[X|Y], E[X|\bar{Y}]\}$ (*)

Let's take MAX-CUT and its 2^n 2-colorings

$$\frac{|E|}{2} = E[\overbrace{E(S, \bar{S})}^{\text{cutsizes}}] \stackrel{(*)}{\leq} E[\text{cutsizes} \mid \text{node 1 is colored according to } (*)]$$

we proved
this at the beginning

in words: first we color node 1 as ● and then we estimate the expected cutsizes given this choice; next we color node 1 as ● and make the corresponding estimation. We choose between ● and ● by taking the maximum of the two choices on node 1

$$\leq E[\text{cutsizes} \mid \text{nodes 1, 2 are colored using } (*)]$$

note that the color of node 1 has been chosen in the previous step, so here we make a choice just for node 2.

...

$$\leq E[\text{cutsizes} \mid \text{nodes 1, 2, ..., } i \text{ are colored using } (*)]$$

we evaluated $2 \cdot i$ choices instead of 2^i . Hence,
we reduced the number of 2-colorings from 2^n to $O(2^{n-i})$

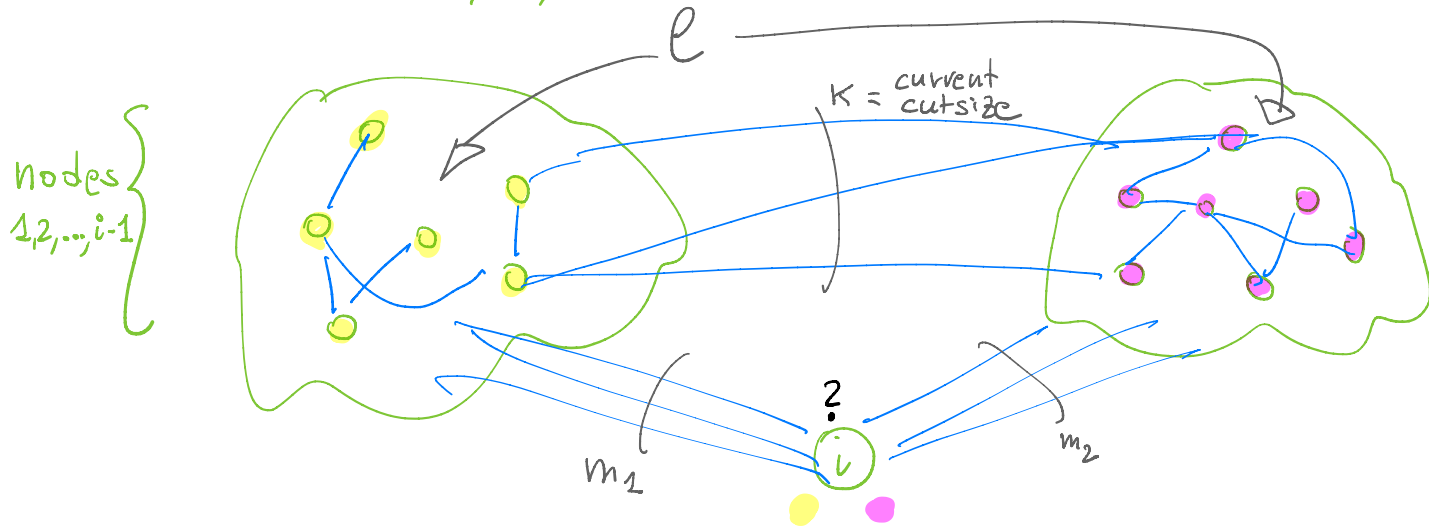
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$$\leq E[\text{cutsizes} \mid \text{all nodes are colored using } (*)]$$

Let's see how to evaluate each step with (*) $E[X] \leq \max \{E[X|Y], E[X|\bar{Y}]\}$

Step i : we have to decide "conditionally" the color for node i :

- nodes $1, 2, \dots, i-1$ are already colored
- nodes $i+1, \dots, n$ will be colored later



Edges in E are grouped consequently $\Rightarrow |E| = m_1 + m_2 + l + k + \underbrace{\quad}_{\text{the remaining edges}}$

$$L = |E| - m_1 - m_2 - k - c$$

We are evaluating over
all possible 2-colourings of nodes $i+1, \dots, n$
↓

$$\textcircled{A} \ E[\text{cutsizes}] \text{ nodes } 1, 2, \dots, i-1 \text{ colored, node } i \text{ (yellow)} = \frac{L}{2} + k + \underline{m_2}$$

$$\textcircled{B} \ E[\text{cutsizes}] \text{ nodes } 1, 2, \dots, i-1 \text{ colored, node } i \text{ (pink)} = \frac{L}{2} + k + \underline{m_1}$$

$$\textcircled{A} > \textcircled{B} \Leftrightarrow \cancel{\frac{L}{2}} + \cancel{k} + m_2 > \cancel{\frac{L}{2}} + \cancel{k} + m_1 \Leftrightarrow m_2 > m_1 \quad !$$

⌚
We obtained the GREEDY
condition in an alternative way! ▶